Tutorial 10: Excel Manipulation and Spreadsheet Modelling for Future Actuaries

(STUDENT VERSION)

Theme: Excel Formulas, Assumption Tables, Expected Value, Present Value and Model Checking

Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

A. Warm-up: From Calculator Thinking to Spreadsheet Modelling

Synergy & Learning Flow

Purpose of this section: This opening section shifts you from using Excel as a calculator to using Excel as a modelling environment. You will see that a future actuary does not only calculate answers. A future actuary builds models that can be checked, changed, explained and reused.

Key Definition / Result

A spreadsheet model is not just a table of numbers. A good spreadsheet model usually contains four connected parts:

$$\text{Inputs / Assumptions} \longrightarrow \text{Calculations} \longrightarrow \text{Outputs} \longrightarrow \text{Checks.}$$

For actuarial-style thinking, the spreadsheet should help us answer questions such as:

- What assumptions are being used?
- Which values are entered manually?
- Which values are calculated by formulas?
- What happens if an assumption changes?
- Can another person check the formulas?
- Can we explain the result in plain language?

Investigation / Excel Mission: Classify Spreadsheet Cells ⋆ Core

Task. You are given a simple spreadsheet with the following cells:

Cell	Value / Formula	Possible Role
B2	Interest rate = 3%	Input / assumption
B3	Review threshold = 500	Input / assumption
A7:D12	Data table	Data / scenario information
E7	`=C7*D7`	Calculation
E13	`=SUM(E7:E12)`	Output summary
F7	`=IF(E7>B3,"Review","OK")`	Check / flag

Reflection Question: Spreadsheet Mindset

Why is a spreadsheet with clear inputs, calculations, outputs and checks more reliable than a spreadsheet where numbers are typed everywhere?

Suggested AI Prompt: Consulting Gemini on Spreadsheet Design ⋆ Core

"Hi Gemini. I need to build an Excel spreadsheet to track financial risks. Act like a professional risk manager and explain to me why I must separate my inputs, calculations, outputs, and checks into different sections instead of typing numbers everywhere."

Common Mistake / Debugging Warning

You may think that if Excel gives a number, the number must be correct. This is not true. Excel follows the formula written by the user. A wrong formula can produce a precise-looking but wrong answer.

B. Excel Mission 1: Formula Basics, Cell References, and Autofill

Synergy & Learning Flow

Why this section matters: Many spreadsheet errors are not caused by difficult mathematics. They are caused by wrong cell references. You must understand relative references, absolute references, and how to safely replicate formulas.

Key Definition / Result

Every Excel formula begins with an equal sign:

`=`

The basic arithmetic operators are:

- `+` : addition,
- `-` : subtraction,
- `*` : multiplication,
- `/` : division,
- `^` : power / exponentiation.

Relative vs. Absolute: A relative reference such as `B2` changes when a formula is copied. An absolute reference such as `$B$2` stays fixed when a formula is copied.

The Autofill Feature: The small green square at the bottom-right corner of an active cell is the **fill handle**. Instead of typing formulas row by row, you can double-click this handle or drag it down to automatically copy the formula down the entire column.

Investigation / Excel Mission: Calculate Total Cost and Autofill ⋆ Core

Task. Create the table below in Excel and calculate total cost for the first row, then use **Autofill** for the rest.

Item	Quantity	Unit Cost	Total Cost
A	3	20	
B	5	15	
C	8	12	
D	2	50	

Formula in D2:

```
=B2*C2
```

Explanation: Hover the mouse over the bottom-right corner of `D2` until the cursor turns into a black cross. Double-click it. Excel automatically changes the row references and fills the column:

```
=B2*C2
=B3*C3
=B4*C4
=B5*C5
```

Investigation / Excel Mission: Use a Fixed Growth Rate ⋆ Core

Task. Calculate projected amounts using a fixed growth rate.

Input area:

Input	Value
Annual growth rate	4%

Projection table:

Year	Starting Amount	Projected Amount
1	1000	
2	1000	
3	1000	
4	1000	
5	1000	

If the annual growth rate is in cell `B2`, and the first year is in cell `A6`, then a suitable formula is:

```
=B6*(1+$B$2)^A6
```

Common Mistake / Debugging Warning

A very common mistake is writing:

```
=B6*(1+B2)^A6
```

This may look correct in the first row. However, when using Autofill, `B2` may become `B3`, `B4`, and so on. If the growth rate is stored only in `B2`, the copied formulas become wrong.

Reflection Question: Absolute Reference

What types of cells should usually be written with absolute references such as `$B$2`?

Suggested AI Prompt: Asking Gemini about Absolute References ⋆ Core

"Hi Gemini, my Excel formula uses an absolute reference for the interest rate, written as $B$2, to multiply against changing values in a table. Explain to me exactly what the dollar signs do in Excel and what happens when I Autofill this formula

down the column without them."

C. Excel Mission 2: Clean Model Layout, No Hard-coding, and Named Ranges

Synergy & Learning Flow

Why this section matters: An actuarial-style spreadsheet should be changeable. If assumptions are hidden inside formulas, the model becomes difficult to update and difficult to audit. Furthermore, using "Named Ranges" elevates a spreadsheet from a grid of confusing letters and numbers to readable math.

Key Definition / Result

Hard-coding means typing an assumption directly into a formula.

Poor modelling style:

```
=1000*(1+0.03)^5
```

Better modelling style (Absolute Reference):

```
=$B$2*(1+$B$3)^$B$4
```

Best modelling style (Named Range):

```
=Starting_Amount*(1+Interest_Rate)^Years
```

How to Name a Range: Click on a cell (like `B3`), go to the **Name Box** (top left, next to the formula bar), type a name without spaces (like `Interest_Rate`), and press Enter.

Investigation / Excel Mission: Improve a Messy Spreadsheet and Add Named Ranges

⋆ **Core**

Task. You are shown this poor spreadsheet layout:

Name	Amount	Rate	Answer
A	1000	0.03	1030
B	2000	0.03	2060
C	1500	0.03	1545

Redesign it into a clean layout:

1. Create a visible input area.
2. Click the cell containing `3%` and name it `Growth_Rate` in the Name Box.
3. Click the cell containing `2000` and name it `Review_Threshold`.
4. Rewrite the formulas using these English names instead of cell coordinates.

Suggested formula using Named Ranges:

```
=B7*(1+Growth_Rate)
```

Suggested flag using Named Ranges:

```
=IF(D7>Review_Threshold,"Review","OK")
```

Common Mistake / Debugging Warning

Do not believe that formatting is only decoration. In modelling, formatting helps the reader distinguish inputs, formulas, outputs and checks. It reduces the chance of changing the wrong cell.

Reflection Question: Visible Assumptions

Why is it dangerous to hide assumptions such as interest rates or probabilities directly inside formulas?

Suggested AI Prompt: Gemini as a Code Reviewer for Hard-coding ⋆ Core

"Gemini, here is my Excel formula for projecting an amount: =1000(1+0.03)^5. I typed the 0.03 directly into the formula. Tell me why 'hard-coding' assumptions like this is a bad habit in professional modelling, and suggest a better way to structure this using Excel cell references or Named Ranges."*

D. Excel Mission 3: The Data Type Trap (Strings vs. Numbers)

Synergy & Learning Flow

Why this section matters: Before learning lookup formulas, you must understand data types. A common, frustrating spreadsheet error occurs when numbers are accidentally formatted or imported as text strings. To Excel, the text string "100" is completely different from the number 100.

Key Definition / Result

In Excel, data is classified by **type**. The two most common types are:

- **Number:** Used for calculations (e.g., `100`). Automatically aligns to the right of the cell.
- **String (Text):** Treated as characters (e.g., `"100"` or `'100`). Automatically aligns to the left of the cell.

You can check a cell's type using logical functions:

```
=ISNUMBER(A2)
=ISTEXT(A2)
```

If you try to use a String "100" to look up the Number 100, Excel will fail and return a `#N/A` error.

Investigation / Excel Mission: Detecting the String vs. Number Trap ⋆⋆

Intermediate

Task. Enter the number `500` into cell `A2`. In cell `A3`, type an apostrophe followed by 500 (i.e., `'500`). The apostrophe forces Excel to store it as a string.

In cells `B2` and `B3`, write:

```
=ISNUMBER(A2)
=ISNUMBER(A3)
```

Then, in cell `C2`, test if they are equal:

```
=A2=A3
```

Common Mistake / Debugging Warning

When downloading financial data from external systems, IDs, years, or currency values often arrive as text strings. Attempting to SUM them or XLOOKUP them will fail. You can fix strings by multiplying them by 1 (`=A3*1`) or using the `VALUE()` function (`=VALUE(A3)`).

Suggested AI Prompt: Consulting Gemini on the Text vs. Number Trap

⋆ **Core**

"Gemini, I have an Excel spreadsheet where the number 500 is in cell A2 and the text string '500' is in cell A3. They look exactly the same. Why does my VLOOKUP or XLOOKUP return an #N/A error when trying to match them, and what are two easy ways to convert text numbers back into real numbers?"

E. Excel Mission 4: Assumption Tables, Data Validation, and `XLOOKUP`

Synergy & Learning Flow

Why this section matters: Actuarial models often depend on assumption tables. You should learn to retrieve assumptions using formulas rather than manually copying values. Furthermore, to prevent lookups from failing due to spelling mistakes, actuaries restrict user inputs using Data Validation.

Key Definition / Result

Data Validation (Dropdown Lists): Restricts what a user can type into a cell to prevent errors like typing "Meduim" or "High " (with an extra space).

- Select cell → **Data** tab → **Data Validation** → Allow: **List** → Source: `Low, Medium, High`.

XLOOKUP: Used to search for a value in one range and return a matching value from another range.

```
=XLOOKUP(value_to_find, lookup_column, return_column)
```

In modelling language:

Find the category ⟶ Return the matching assumption.

Investigation / Excel Mission: Prevent Lookup Errors with Data Validation ⋆

Core

Task. Before writing any formulas, create a dropdown menu for the "Risk Level" column in your main table to ensure the spelling matches the assumption table exactly.

1. Highlight the Risk Level cells in the main table (`B7:B10`).
2. Click **Data Validation**.
3. Choose **List** and type `Low, Medium, High`.

Investigation / Excel Mission: Retrieve Risk Probabilities ⋆ Core

Task. Create an assumption table:

Risk Level	Probability
Low	5%
Medium	15%
High	30%

Then fill in your Data Validated main table:

Item	Risk Level (Dropdown)	Cost if Occurs	Probability
A	Low	1000	
B	High	2500	
C	Medium	1800	
D	High	3000	

If the risk level in the main table is in `B7`, the assumption table risk levels are in `G3:G5`, and the probabilities are in `H3:H5`, then the formula is:

```
=XLOOKUP(B7,$G$3:$G$5,$H$3:$H$5)
```

Investigation / Excel Mission: Calculate Expected Cost ⋆ Core

Task. After retrieving the probability, calculate expected cost.

$$\text{Expected Cost} = \text{Cost if Occurs} \times \text{Probability}.$$

Formula example:

```
=C7*D7
```

Common Mistake / Debugging Warning

You might be tempted to manually copy probabilities into the main table. For a very small table, this may seem harmless. However, in a larger model, manual copying creates inconsistency risk. If an assumption changes, the copied values may not update.

Suggested AI Prompt: Explaining `XLOOKUP` for Assumption Tables ⋆ **Core**

"Explain XLOOKUP to a secondary school student using a risk level table. The main table contains Low, Medium and High risk items. The assumption table contains matching probabilities. Please explain why lookup formulas are safer than manually copying probabilities."

Reflection Question: Linked Assumptions

Why is it safer to link assumptions from one table rather than typing the same assumptions repeatedly?

F. Excel Mission 5: Logical Formulas, Warning Flags, and Conditional Formatting

Synergy & Learning Flow

Why this section matters: A good model should not only calculate results. It should also warn the user when something may be wrong or unusual, and visual flags make these warnings impossible to miss.

Key Definition / Result

The `IF` function creates a simple decision rule:

```
=IF(condition, value_if_true, value_if_false)
```

Examples:

```
=IF(A2>1000,"Review","OK")
=IF(B2<0,"Check negative","OK")
=IF(C2="","Missing","OK")
```

Conditional Formatting: You can make Excel automatically change cell colors based on values. For instance, making the background red if the cell contains the word "Review".

Investigation / Excel Mission: Create Review Flags and Apply Formatting ⋆ Core

Task. Add a flag column to the risk-cost model.

Suppose the expected cost is in `E7`, and the review threshold is stored in a cell named `Review_Threshold`. The formula is:

```
=IF(E7>Review_Threshold,"Review","OK")
```

Copy the formula down. Once the column is filled, apply **Conditional Formatting**:

1. Highlight the flag column.
2. Go to **Home** → **Conditional Formatting** → **Highlight Cells Rules** → **Equal To...**
3. Type `Review` and select "Light Red Fill with Dark Red Text".

Investigation / Excel Mission: Check Probability Validity ⋆⋆ Intermediate

Task. Create a check to ensure that probabilities are between 0 and 1.

Formula example:

```
=IF(AND(D7>=0,D7<=1),"OK","Invalid probability")
```

Investigation / Excel Mission: Check Missing Values ⋆ Core

Task. Create a check for missing risk level.

Formula example:

```
=IF(B7="","Missing risk level","OK")
```

Suggested AI Prompt: Debugging an `IF` Formula ⋆ Core

"Here is my Excel IF formula. Please check whether the condition, true result and false result are in the correct order. Explain the formula in simple words and sug-

gest a corrected version if needed."

Reflection Question: Model Warnings

Why is it useful for a spreadsheet to use conditional formatting to highlight words like "Review" or "Invalid" rather than just presenting a wall of black-and-white text?

G. Excel Mission 6: Conditional Summaries with `SUMIFS` and `COUNTIFS`

Synergy & Learning Flow

Why this section matters: Individual row calculations are useful, but future actuaries also need to summarise results by categories. Conditional summary formulas turn detailed row-level outputs into decision-level information.

Key Definition / Result

`SUMIFS` adds values that satisfy one or more conditions:

```
=SUMIFS(sum_range, criteria_range1, criteria1, criteria_range2, criteria2)
```

`COUNTIFS` counts entries that satisfy one or more conditions:

```
=COUNTIFS(criteria_range1, criteria1, criteria_range2, criteria2)
```

Investigation / Excel Mission: Summarise Expected Cost by Risk Level

⋆⋆ **Intermediate**

Task. Use your main risk table to answer summary questions.

Example table:

Item	Team	Risk Level	Expected Cost
A	Team 1	High	2000
B	Team 2	Low	300
C	Team 1	Medium	800
D	Team 1	High	1500
E	Team 2	High	1800

Total expected cost for Team 1:

```
=SUMIFS(D2:D6,B2:B6,"Team 1")
```

Number of high-risk items:

```
=COUNTIFS(C2:C6,"High")
```

Total expected cost for high-risk items in Team 1:

```
=SUMIFS(D2:D6,B2:B6,"Team 1",C2:C6,"High")
```

Common Mistake / Debugging Warning

For `SUMIFS`, the first argument is the range to be summed. It is easy to confuse this with `SUMIF`, where the order of arguments is different. In this tutorial, use `SUMIFS` consistently to avoid switching structures.

Investigation / Excel Mission: Build a Summary Panel ⋆⋆ Intermediate

Task. Create a summary panel with the following questions:

Summary Question	Formula Type
Total expected cost for high-risk items	`SUMIFS`
Total expected cost for Team 1	`SUMIFS`
Number of items marked Review	`COUNTIFS`
Number of invalid probabilities	`COUNTIFS`

Suggested AI Prompt: Troubleshooting SUMIFS with Gemini ⋆⋆ Intermediate

"Gemini, I am trying to use the SUMIFS function in Excel to find the total expected cost for 'High' risk items in 'Team 1', but I am getting a #VALUE! error or an answer of 0. What are the most common mistakes people make with the order of arguments or range sizes in SUMIFS?"

Reflection Question: From Rows to Decisions

Why is a summary panel useful when a spreadsheet contains many rows of calculations?

H. Excel Mission 7: Expected Value, SUMPRODUCT, and Expected Present Value

Synergy & Learning Flow

Why this section matters: This is the conceptual centre of the tutorial. You will learn the fundamental risk pattern of Expected Value, and how `SUMPRODUCT` can calculate it efficiently. Then, we add timing to form Expected Present Value.

Key Definition / Result

Expected value:

$$\text{Expected Value} = \text{Amount} \times \text{Probability}.$$

SUMPRODUCT shortcut: Instead of creating a helper column to multiply Amount and Probability row-by-row and then summing it, `SUMPRODUCT` does it in one step:

```
=SUMPRODUCT(Amount_Range, Probability_Range)
```

Discount factor:

$$\text{Discount Factor} = \frac{1}{(1+r)^t}.$$

Present value & Expected present value:

$$\text{PV} = \text{Future Amount} \times \text{Discount Factor}.$$

$$\text{EPV} = \text{Future Amount} \times \text{Probability} \times \text{Discount Factor}.$$

Investigation / Excel Mission: Expected Value Table and `SUMPRODUCT` ⋆ Core

Task. Calculate expected amounts for possible uncertain payments.

Scenario	Amount (B)	Probability (C)	Expected Amount (D)
A (Row 2)	1000	10%	
B (Row 3)	500	40%	
C (Row 4)	2000	5%	
D (Row 5)	800	25%	

Step 1: Row-by-Row Expected Amount (in D2):

```
=B2*C2
```

Sum the column: `=SUM(D2:D5)`

Step 2: The Actuary's Shortcut: In a new cell, use `SUMPRODUCT` to get the total Expected Value without needing column D.

```
=SUMPRODUCT(B2:B5, C2:C5)
```

Investigation / Excel Mission: Discount Factor and Expected Present Value

⋆⋆ **Intermediate**

Task. Calculate discount factors and EPV.

Scenario	Year	Future Amount	Probability	Discount Factor	EPV
A	1	1000	10%		
B	2	500	40%		
C	3	2000	5%		
D	4	800	25%		

Suppose the interest rate is stored in cell `B1` (or named `Interest_Rate`).

Discount factor formula:

```
=1/(1+Interest_Rate)^B2
```

Expected present value formula:

```
=C2*D2*E2
```

Suggested AI Prompt: Gemini to Verify Conceptual Understanding

⋆⋆ **Intermediate**

*"Gemini, I am studying expected present value (EPV) for actuarial models. My Excel formula is EPV = Future Amount * Probability * Discount Factor. Can you explain to me, like I am 15 years old, why we need to multiply by BOTH a probability percentage and a discount factor when evaluating future risk? Also briefly mention why actuaries love using SUMPRODUCT."*

Reflection Question: Expected Present Value

Why does expected present value combine amount, probability and timing?

I. Excel Mission 8: Actuarial Pricing and Goal Seek

Synergy & Learning Flow

Why this section matters: Once actuaries calculate the Expected Cost of a risk portfolio, they must make a business decision: "What Premium do we charge the customer to cover this cost?" Excel's Goal Seek feature allows you to calculate pricing instantly.

Key Definition / Result

Goal Seek (What-If Analysis): An Excel tool that mathematically works backward. You tell Excel what final result you want, and it will change an input cell until it finds the exact number needed to achieve that result.

Basic Pricing Formula:

$$\text{Profit} = (\text{Premium per Client} \times \text{Number of Clients}) - \text{Total Expected Cost}$$

Investigation / Excel Mission: Calculate the Break-Even Premium ⋆⋆

Intermediate

Task. Set up a pricing dashboard to figure out what to charge 50 clients to cover a total risk pool of $5,000.

Pricing Input	**Value**
Total Expected Cost	5000
Number of Clients	50
Premium per Client	**0** *(Leave blank for now)*
Total Profit	*Formula here*

Formula for Profit (assuming cells B2, B3, B4):

```
=(B4*B3)-B2
```

Use Goal Seek:

1. Go to **Data** → **What-If Analysis** → **Goal Seek**.
2. **Set cell:** Click the **Profit** cell.
3. **To value:** Type `0` (for break-even) or `500` (for a profit margin).
4. **By changing cell:** Click the **Premium per Client** cell.

Suggested AI Prompt: Gemini on Goal Seek and Pricing ⋆⋆ **Intermediate**

"Gemini, I just used Excel's 'Goal Seek' to find the exact Premium I need to charge 50 clients to cover an Expected Cost of $5000. Explain to me how Goal Seek works behind the scenes, and why finding a break-even price is so important for an insurance company."

J. Excel Mission 9: The Law of Large Numbers and Simulation

Synergy & Learning Flow

Why this section matters: Expected Value is a deterministic, flat average. But real life is chaotic. Sometimes a 10% risk happens, sometimes it doesn't. By using Excel to simulate randomness, you can visualize the "Law of Large Numbers" and see why actuaries need thousands of policies to be safe.

Key Definition / Result

Stochastic Simulation: Using random numbers to see what "might" happen.

- `=RANDBETWEEN(1, 100)` generates a random whole number between 1 and 100.
- Pressing **F9** (or Formulas → Calculate Now) forces Excel to recalculate and generate a new random number.

Investigation / Excel Mission: Simulating a Risk Event ⋆⋆⋆ **Challenge**

Task. An event has a **10% probability** of occurring and costs $1000 if it does. The Expected Value is $100. Let's see what happens in reality.

In a new cell, write this simulation formula:

```
=IF(RANDBETWEEN(1,100)<=10, 1000, 0)
```

The Experiment:

1. Press **F9** rapidly 10 to 15 times to recalculate.
2. Observe the cell. Most of the time it says `0`. Occasionally, it flashes `1000`.
3. Notice that it **never** says `100` (the Expected Value).

Reflection Question: Expected Value vs. Reality

Why is the Expected Value (\$100) not the same as the actual realised outcome (\$0 or \$1000)?

K. Excel Mission 10: Scenario Thinking, Sensitivity, and Visualising Risk

Synergy & Learning Flow

Why this section matters: Actuarial models depend on assumptions. You will see that changing an assumption can change the output. Furthermore, actuaries must communicate complex math to executives; visualising risk with a chart is far more effective than a table of numbers.

Key Definition / Result

A **sensitivity test** asks:

What happens to the output if one assumption changes?

Examples:

- What if the interest rate increases?
- What if the probability estimate is higher?
- What if the future amount is larger?

Line Charts: Connecting data points with a line to visually demonstrate a trend or sensitivity slope.

Investigation / Excel Mission: Interest Rate Sensitivity and Visualisation

⋆⋆ **Intermediate**

Task. Compare expected present value under three interest-rate scenarios and plot a chart.

Scenario	Interest Rate	Calculated EPV
Low-rate	2%	*(Calculate)*
Base-rate	3%	*(Calculate)*
High-rate	5%	*(Calculate)*

Create the Chart:

1. Highlight the Interest Rate and EPV columns.
2. Go to **Insert** → **Charts** → **Insert Line or Area Chart** → **Line with Markers**.

Discussion prompt:

When interest rate increases, what visually happens to the expected present value on the chart?

Investigation / Excel Mission: Probability Sensitivity ⋆⋆ Intermediate

Task. Increase each probability by 20% proportionally and compare the new expected value.

If the original probability is in C2, then the stressed probability may be:

```
=C2*1.2
```

A safer version with a cap at 100% is:

```
=MIN(C2*1.2,1)
```

Suggested AI Prompt: Explaining Sensitivity Testing ⋆⋆ Intermediate

"I changed the interest rate and probabilities in my Excel expected present value model and plotted a line chart. Help me explain how changing assumptions affects the output. Please explain in beginner-friendly language and remind me that sensitivity testing is not the same as predicting the future."

Reflection Question: Assumptions Drive Outputs

Why should an actuary be careful when choosing assumptions for a spreadsheet model?

L. Excel Mission 11: Model Checks and Error Detection

Synergy & Learning Flow

Why this section matters: A spreadsheet can look professional but still contain hidden errors. You will learn to build simple checks directly into the model.

Key Definition / Result

A model check is a formula that helps detect possible problems.

Common checks include:

- probability must be between 0 and 1;
- amount should not be negative unless a negative value is meaningful;
- lookup result should not be missing;
- totals should reconcile;
- output should not be blank;
- formulas should be copied consistently.

Investigation / Excel Mission: Probability Check ⋆ Core

Task. Check whether a probability is valid.

```
=IF(AND(C2>=0,C2<=1),"OK","Check probability")
```

Investigation / Excel Mission: Lookup Check ⋆⋆ Intermediate

Task. Check whether the lookup result is available.

One possible formula is:

```
=IFERROR(XLOOKUP(B2,$G$2:$G$4,$H$2:$H$4),"Missing assumption")
```

Another approach is to keep the lookup formula and create a separate check:

```
=IF(ISNUMBER(D2),"OK","Check lookup")
```

Investigation / Excel Mission: Reconciliation Check ⋆⋆ **Intermediate**

Task. Check whether a reported total equals the sum of components.

```
=IF(ABS(B10-SUM(B2:B9))<0.01,"OK","Check total")
```

Common Mistake / Debugging Warning

Do not use `IFERROR` to hide every error. If a formula produces an error, the modeller should first understand why. Error handling should support review, not cover up mistakes.

Suggested AI Prompt: Auditing My Spreadsheet Model ⋆⋆ **Intermediate**

"I built an Excel model with inputs, formulas, lookup formulas, expected present value calculations and checks. Please help me review possible spreadsheet errors. Focus on wrong cell references, string vs. number mismatches, missing assumptions, invalid probabilities and unclear formula logic."

Reflection Question: Trusting a Model

What makes a spreadsheet result trustworthy?

M. Integrated Excel Challenge: Build a Simple Risk Cost Model

Synergy & Learning Flow

Purpose of this challenge: You will integrate the skills from the whole tutorial.

Investigation / Excel Mission: Build the Model Structure ⋆⋆ **Intermediate**

Task. Build a spreadsheet with the following parts:

1. Input section (Using **Named Ranges**);
2. Risk probability assumption table;
3. Main item table (Using **Data Validation** dropdowns for Risk Level);
4. Lookup formulas;

5. Discount factor calculation;
6. Expected present value calculation;
7. Review flag (with **Conditional Formatting**);
8. Summary panel (using **SUMPRODUCT** and **SUMIFS**);
9. Pricing Dashboard (using **Goal Seek**);
10. Model checks;
11. Written interpretation.

Suggested input section:

Input	Value
Interest rate	3%
Review threshold	500

Suggested assumption table:

Risk Level	Probability
Low	5%
Medium	15%
High	30%

Investigation / Excel Mission: Complete the Main Table ⋆⋆ Intermediate

Main table:

Item	Year	Risk	Future Cost	Prob.	DF	EPV	Flag
A	1	Low	1000				
B	2	High	2500				
C	3	Medium	1800				
D	4	High	3000				
E	5	Low	1200				

Probability lookup:

```
=XLOOKUP(C7,$J$3:$J$5,$K$3:$K$5)
```

Discount factor:

```
=1/(1+Interest_Rate)^B7
```

Expected present value:

```
=D7*E7*F7
```

Review flag:

```
=IF(G7>Review_Threshold,"Review","OK")
```

Investigation / Excel Mission: Build the Summary Panel ⋆⋆ Intermediate

Task. Build a summary panel.

Total expected present value:

```
=SUM(G7:G11)
```

Number of review items:

```
=COUNTIFS(H7:H11,"Review")
```

Total EPV for high-risk items:

```
=SUMIFS(G7:G11,C7:C11,"High")
```

Number of high-risk items:

```
=COUNTIFS(C7:C11,"High")
```

Investigation / Excel Mission: Add Model Checks ⋆⋆ Intermediate

Task. Add at least three checks.

Probability validity check:

```
=IF(AND(E7>=0,E7<=1),"OK","Check probability")
```

Missing risk level check:

```
=IF(C7="","Missing risk level","OK")
```

Negative future cost check:

```
=IF(D7<0,"Check negative cost","OK")
```

Reflection Question: Integrated Model

Which part of the model is most important for making it reliable: inputs, formulas, outputs or checks? Explain your choice.

Suggested AI Prompt: Gemini as a Final Project Co-Pilot ⋆⋆⋆ **Challenge**

"Gemini, I have just built a complete risk cost model in Excel with an input section, lookup formulas, EPV calculations, and summary flags. As an expert risk manager, what are the top 3 stress tests or scenario checks I should manually perform on my spreadsheet to guarantee the formulas are bulletproof before I present it?"

N. AI-supported Excel Investigation Protocol

Synergy & Learning Flow

How you should use AI in this tutorial: AI can help explain formulas, identify wrong references, debug lookup errors and improve interpretation. However, you should not use AI simply to produce final answers. You must understand the formula logic and check whether the AI explanation makes sense.

Key Definition / Result

A good AI-assisted Excel workflow:

1. Build the spreadsheet structure yourself.
2. Write or attempt the formula yourself.
3. Observe the output or error.
4. Ask AI to explain the formula or error.
5. Check whether the AI explanation agrees with the modelling logic.
6. Correct the formula if needed.
7. Write an interpretation in your own words.

Suggested AI Prompt: General Excel Formula Debugging ⋆ **Core**

"I am learning Excel spreadsheet modelling. Here is my formula and the result I obtained. Please explain what the formula is doing, identify possible reference errors, and suggest a corrected version if needed. Please explain in beginner-friendly language."

Suggested AI Prompt: Checking an `XLOOKUP` Formula ⋆ **Core**

"I am using XLOOKUP to retrieve a probability from an assumption table. Here is my formula. Please check whether the lookup value, lookup range and return range are correct. Also explain why some ranges may need dollar signs."

Common Mistake / Debugging Warning

Do not ask AI only for the final answer. Ask AI to explain your formula, challenge your assumptions and help you find possible errors. You are still responsible for the modelling logic.

O. Consolidation and Exit Questions

Investigation / Excel Mission: Core Excel Skills Review ⋆ **Core**

1. What symbol begins an Excel formula?
2. What is the difference between `B2` and `$B$2`?
3. How does the Autofill feature save time, and where do you click to use it?
4. Why are Named Ranges (like `Interest_Rate`) better than hard-coded numbers or absolute references?
5. Why does Excel treat the number `500` and the string `"500"` differently?

Investigation / Excel Mission: Lookup, Validation, and Logic Review ⋆ **Core**

1. What does `XLOOKUP` do?
2. How does Data Validation (a dropdown list) prevent `XLOOKUP` errors?
3. What does `IF` do?
4. Give one example of a useful model flag.
5. How does Conditional Formatting improve a spreadsheet with many rows?

Investigation / Excel Mission: Expected Value, Pricing, and Simulation Review ⋆⋆ **Intermediate**

1. What does the `SUMPRODUCT` formula calculate?
2. What three ideas are combined in expected present value?
3. How does Excel's Goal Seek help an actuary figure out the right Premium to charge?
4. When using `RANDBETWEEN` to simulate a 10% event, why did the actual cost jump between \$0 and \$1000 instead of sitting safely at the \$100 Expected Value?
5. If the interest rate increases, what visually happens to the present value line on a Sensitivity Chart?

Investigation / Excel Mission: Conditional Summary Review ⋆⋆ **Intermediate**

1. What does `SUMIFS` calculate?
2. What does `COUNTIFS` calculate?
3. Why are summary panels useful?
4. What is one common mistake when using `SUMIFS`?
5. Why should the criteria range and sum range have matching sizes?

Investigation / Excel Mission: AI-supported Learning Review ⋆⋆ **Intermediate**

1. How can AI help when an Excel formula gives an error?
2. Why should we not blindly trust AI-generated formulas?
3. What should we check after AI explains an `XLOOKUP` formula?
4. How can AI help us explain expected present value?
5. What is one good AI prompt you used or would use in this tutorial?

Investigation / Excel Mission: Challenge Questions ⋆⋆⋆ Challenge

1. If an assumption changes, why should a well-built spreadsheet update automatically?
2. Why is a model with many hard-coded values difficult to audit?
3. If a model has a correct formula but a wrong assumption, can the output still be misleading?
4. Why is sensitivity testing useful?
5. Why is expected value not the same as the actual realised outcome?
6. Why do we need both formula checks and human interpretation?
7. How would you explain expected present value to a younger student?
8. What makes an Excel model actuarial in spirit, even if it uses simple examples?

Key Definition / Result

Final Takeaway. This tutorial changes the focus from programming scripts to spreadsheet modelling. Excel helps us organise assumptions, restrict inputs with validation, calculate values, connect tables, simulate reality, visualise scenarios, and check errors.

The most important modelling pattern is:

$$\text{Expected Present Value} = \text{Amount} \times \text{Probability} \times \text{Discount Factor}.$$

The most important spreadsheet habit is:

Do not only calculate. Build a model that can be checked, changed and explained.

AI can support the learning process by explaining formulas, debugging errors and helping interpretation, but you must still understand the assumptions, formulas, outputs and limitations.

End of Tutorial 10